

ISLA MAGUEYES, Puerto Rico

WMO: 997350

Lat: 17.9700N

Lon: 67.0464W

Elev: 16

StdP: 14.69

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	68.9	70.0	N/A	N/A	N/A	N/A	N/A	N/A	17.8	80.9	14.7	80.5	3.4	50	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.5	88.6	N/A	88.0	N/A	87.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.0	11.0	DB	66.6	90.4	1.1	1.0	65.8	91.1	65.1	91.8	64.5	92.3	63.8	93.1	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.1	77.3	77.1	77.0	78.8	80.6	82.4	82.4	82.6	82.4	81.9	80.2	78.4
	DBStd	2.66	1.39	1.55	2.01	1.75	1.63	1.14	1.26	1.33	1.39	1.49	1.59	1.76
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10989	846	758	838	863	948	971	1006	1010	972	990	905	882
	CDD65	5514	381	338	373	413	483	521	541	545	522	525	455	417
	CDH74	55081	2770	2500	2804	3677	4993	6106	6340	6434	5987	5780	4291	3399
	CDH80	15767	436	347	372	734	1331	2059	2242	2316	2077	1905	1187	761
Wind	WSAvg	4.4	3.9	4.6	5.0	5.1	4.7	4.7	4.5	4.6	4.3	4.3	3.9	3.9
Precipitation	PrecAvg	60.5	3.3	2.4	2.8	4.8	6.1	4.4	5.2	6.6	7.4	6.9	6.5	4.0
	PrecMax	90.5	7.8	5.4	5.9	12.2	11.9	9.4	10.6	12.2	17.7	17.0	17.4	8.9
	PrecMin	32.9	1.1	1.2	0.3	0.9	2.5	1.5	2.4	3.0	2.3	3.0	1.2	1.2
	PrecStd	14.9	1.7	1.0	1.5	2.8	2.9	2.0	1.9	1.8	3.6	3.3	3.1	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.6	84.2	85.1	85.9	87.8	88.5	89.0	89.1	89.0	89.0	87.4	86.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	83.7	83.4	83.9	84.9	86.4	87.7	88.1	88.2	88.1	88.0	86.8	85.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	83.1	82.7	82.9	84.2	85.5	87.0	87.4	87.7	87.4	87.2	86.0	84.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	82.4	82.1	82.1	83.4	84.9	86.4	86.9	87.0	86.8	86.5	85.2	83.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	10.9	10.7	9.9	9.4	8.6	8.5	9.4	9.5	9.5	9.6	10.5	11.1
		MCDBR	11.4	10.8	9.8	9.5	8.8	9.0	9.8	9.8	9.8	10.0	10.8	11.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.372	0.392	0.389	0.417	0.457	0.530	0.524	0.488	0.436	0.391	0.386	0.382	
	taud	2.551	2.480	2.493	2.435	2.337	2.092	2.109	2.241	2.421	2.558	2.572	2.554	
	Ebn at Noon	290	289	292	283	268	246	249	260	274	285	283	283	
	Edh at Noon	31	35	35	37	41	52	51	45	37	32	30	30	
All-Sky Solar Radiation	RadAvg	1519	1765	1965	2055	1983	1993	2017	1978	1860	1665	1470	1427	
	RadStd	49	37	74	98	128	108	91	74	131	120	112	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.59	+1.06	N/A	N/A	N/A	N/A	N/A	N/A	+203	+202

Nomenclature: See separate page